S Haresh

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.33
ROYAL INT SCH PALLAKKAPALAYAM NAMAKKAL TN (55261)	Class XII (CBSE)	2024	97.40%
ROYAL INT SCH PALLAKKAPALAYAM NAMAKKAL TN (55261)	Class X (CBSE)	2022	97.00%

PROJECTS

Live Facial Emotion Classifier: Git

PyTorch, CNN, FastAPI, Docker, OpenCV, Render, JavaScript

- Designed and trained a convolutional neural network (CNN) from scratch in PyTorch (no pretrained backbone) on the FER2013 dataset, achieving ~60% validation accuracy on a 7-class facial emotion recognition task -- in line with published benchmarks for non-pretrained models on this notoriously difficult, noisily-labeled dataset.
- Diagnosed a train/inference distribution mismatch causing degraded real-world performance -- webcam frames were far less tightly cropped than training data -- and resolved it by integrating OpenCV face-detection preprocessing, yielding a substantial accuracy improvement in live testing.
- Engineered a full deployment pipeline: containerized the model with Docker, served predictions through a FastAPI backend, and deployed the service on Render with a live, browser-based webcam demo performing real-time inference.
- Built a responsive, privacy-conscious frontend in JavaScript, client-side image capture, a session gallery with full-size preview, and mobile-responsive layouts -- with all captured images stored locally in the browser. Website

Ask My Docs -- RAG + Redis Q&A System: Git

Python, Redis, RAG, Sentence-Transformers, Ollama, Vector Search (HNSW), Docker

- Designed and implemented a Retrieval-Augmented Generation pipeline from scratch, combining local sentence-transformer embeddings with Redis Stack to perform HNSW-based approximate nearest-neighbor vector search over chunked document embeddings, enabling grounded, citation-backed answers over unstructured PDF/text sources.
- Engineered a semantic caching layer using a second Redis vector index over historical queries, checking cosine similarity against past questions before invoking the LLM to eliminate redundant generation calls on previously queried duplicate questions, with a tunable similarity threshold to balance cache-hit rate against false-positive risk.
- Built the system to run entirely locally with zero external API cost: Uses sentence-transformers (local CPU inference) and Ollama (local LLM serving via an OpenAI-compatible API), while structuring the codebase so either component can be swapped back to a hosted provider by changing a single config value.

Next-Word Generation & Language Modeling (NLP)

PyTorch, TensorFlow, Transformer Architectures, N-gram/Markov Models, KV-Caching

- **GPT-Style Transformer from Scratch** -- Implemented a decoder-only transformer in TensorFlow from first principles, including multi-head self-attention, positional encoding, and causal masking, without relying on high-level attention APIs. Engineered a KV-cache to eliminate redundant computation during autoregressive inference, and trained/validated the model on custom text data with checkpointing for reproducibility. Transformer
- **Neural Next-Word Predictor** -- Built a PyTorch-based neural network to predict the next word given a seed sentence, exploring embedding and recurrent layers as a precursor to the transformer implementation. NN_Word_Predictor
- **N-gram/Markov Word Predictor** -- Developed a word-level generator using n-gram probability models to predict the next word in a sequence.

SCHOLASTIC ACHIEVEMENTS

Selected for Sakura Science Program:

Got selected as one among the three participants for the SSP set in Oct, 2026 based on merit (CGPA).

POSITIONS OF RESPONSIBILITY

Domain Head of International Relations Cell (2026-27):

Acting Head of Inbound, Logistics & Campus Tour Domain of International Relations Cell, IITH.

Have contributed to hosting multiple events aimed at international cultural exchange as well as academic exchange among both international and on campus students.

Acting point-of-contact of over 20 Coordinators tasked with maintaining relations with international students.

Class Representative (2026-27) :

Acting CR of Engineering Physics 24' batch.

Point-of-contact between Faculty and Students.

SKILLS

Technical/Softwares: Git, GitHub, HuggingFace, Kaggle, FastAPI, Docker, Render,

Libraries: qiskit, pandas, torch, torchvision, tensorflow, scikit, pillow, opencv, gradio, jax, numpy, fastapi.

RELEVANT COURSES

AI/ML: Computational Physics, Introduction to Probability, Introduction to Statistics. Artificial Intelligence

Coding courses: Introduction to Programming, Data Structures and Applications.

EXTRACURRICULAR

Rubik's Cube (3x3): Average solve time of 15 seconds while fastest time ever being 8.73 seconds.

Gym: Stuck to regime for over a year, reflecting strong habit formation, discipline and time management